

34)d)

| $P$ | $Q$ | $(P \wedge Q)$ | $(P \vee Q)$ | $(P \wedge Q) \rightarrow (P \vee Q)$ |
|-----|-----|----------------|--------------|---------------------------------------|
| T   | T   | T              | T            | T                                     |
| T   | F   | F              | T            | T                                     |
| F   | T   | F              | T            | T                                     |
| F   | F   | F              | F            | T                                     |

e)  $(Q \rightarrow \neg P) \leftrightarrow (P \leftrightarrow \neg Q)$

| $P$ | $Q$ | $P \leftrightarrow \neg Q$ | $\neg P$ | $Q \rightarrow \neg P$ | $(Q \rightarrow \neg P) \leftrightarrow (P \leftrightarrow \neg Q)$ |
|-----|-----|----------------------------|----------|------------------------|---------------------------------------------------------------------|
| T   | T   | T                          | F        | F                      | T                                                                   |
| T   | F   | F                          | F        | T                      | F                                                                   |
| F   | T   | F                          | T        | T                      | F                                                                   |
| F   | F   | T                          | T        | T                      | T                                                                   |

f)  $(P \leftrightarrow \neg Q) \oplus (P \leftrightarrow \neg P)$

| $P$ | $Q$ | $P \leftrightarrow \neg Q$ | $\neg Q$ | $P \leftrightarrow \neg P$ | $(P \leftrightarrow \neg Q) \oplus (P \leftrightarrow \neg P)$ |
|-----|-----|----------------------------|----------|----------------------------|----------------------------------------------------------------|
| T   | T   | T                          | F        | F                          | T                                                              |
| T   | F   | F                          | T        | T                          | T                                                              |
| F   | T   | F                          | F        | T                          | T                                                              |
| F   | F   | T                          | T        | F                          | T                                                              |

35) Construct a truth table for each of these compound propositions

a)  $(P \vee Q) \rightarrow (P \oplus Q)$

| $P$ | $Q$ | $P \vee Q$ | $P \oplus Q$ | $(P \vee Q) \rightarrow (P \oplus Q)$ |
|-----|-----|------------|--------------|---------------------------------------|
| T   | T   | T          | F            | F                                     |
| T   | F   | T          | T            | T                                     |
| F   | T   | T          | T            | T                                     |
| F   | F   | F          | F            | T                                     |

b)  $(P \oplus Q) \rightarrow (P \wedge Q)$

| $P$ | $Q$ | $P \oplus Q$ | $P \wedge Q$ | $(P \oplus Q) \rightarrow (P \wedge Q)$ |
|-----|-----|--------------|--------------|-----------------------------------------|
| T   | T   | F            | T            | T                                       |
| T   | F   | T            | F            | F                                       |
| F   | T   | T            | F            | F                                       |
| F   | F   | F            | F            | T                                       |

c)  $(P \vee Q) \oplus (P \wedge Q)$

| $P$ | $Q$ | $P \vee Q$ | $P \wedge Q$ | $(P \vee Q) \oplus (P \wedge Q)$ |
|-----|-----|------------|--------------|----------------------------------|
| T   | T   | T          | T            | F                                |
| T   | F   | T          | F            | T                                |
| F   | T   | T          | F            | T                                |
| F   | F   | F          | F            | F                                |